Wangjie Su

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Research Interests

Current focus: Reliable AI agents for scientific simulation and industrial workflows, with an emphasis on domain-tool and knowledge integration, long-horizon memory and context, multi-agent orchestration, failure recovery, and evaluation.
Future directions: Grounding long-horizon ReAct agents in multimodal perception, sensor streams, world models, and physical action to build perception–planning–action–feedback loops for autonomous laboratories, scientific instruments, and general-purpose robots.

Education

Peking University, B.S. in Theoretical and Applied Mechanics **Sep 2023 – Present**
Peking University, Double Degree in Intelligent Science and Technology **Sep 2024 – Present**

- **GPA**: 3.818/4.0
- **Selected Coursework**: Data Structures and Algorithms (**100**), Programming in AI (**100**), Computer Vision, Probability and Statistics, Numerical Analysis, Generative Modeling, Intelligent Robotics, Trustworthy ML, LLMs and NLG, and Physics Simulation for Computer Graphics.

Research Experience

AI + Fusion Energy Series Project *Undergraduate Researcher* **Mar 2026 – Present**
Xinao Technology Innovation Center, Peking University | PIs: Prof. Yunfeng Liang and Prof. Songfang Huang

- **Problem:** Developing AI agents that can carry out long-horizon fusion-energy simulation workflows, from task planning and knowledge retrieval to tool execution, result interpretation, and iterative refinement.
- **Implementation:** Using **Nexgent** as the experimental harness to connect domain knowledge and simulation tools through MCP, coordinate specialist agents and explicit workflows, and preserve sessions, checkpoints, permissions, and tool traces.
- **Research focus:** Investigating long-context retrieval and recovery, sustained adherence and progress in long-horizon tasks, and permission-gated failure recovery so scientific runs remain state-consistent, auditable, and reproducible.

Combustion Laboratory, Peking University *Undergraduate Researcher* **May 2025 – Present**
PI: Prof. Zheng Chen

- **Background:** Studying fundamental combustion problems in energy conversion and propulsion, including flame dynamics, chemical kinetics, transport, ignition, flame propagation and stabilization, and laminar flame speed.
- **Current work:** Exploring mathematical formulations of multiscale combustion processes from conservation laws and reaction mechanisms, together with machine-learning approaches to predicting key quantities and approximating complex processes.
- **Methodological focus:** Comparing the regimes of mechanistic and data-driven models, with attention to physical consistency, interpretability, data efficiency, and out-of-distribution generalization toward physics–data integrated combustion modeling.

Work Experience

Hangzhou Yuanxi Technology Co., Ltd. *Independent Project Lead* **Jul 2026 – Present**
Early Technical Team Member | System: ManySelves — An Independently Designed General-Purpose Agent Capability Substrate

- **Capability definition:** Reframed an agent team as a versionable capability specification rather than a fixed application or prompt bundle: Identity contracts define roles, boundaries, I/O, and handoffs; Skill documents encode reusable methods and acceptance criteria; a stable runtime realizes both without domain-specific rewrites.

- **Execution abstraction:** Independently built the system end to end around one Main Agent and task-scoped specialists operating on project-local knowledge, state, and artifacts. Typed state, checkpoints, rollback, resumable decisions, and explicit review boundaries make long-horizon behavior inspectable and recoverable.
- **Research platform:** FastAPI exposes only conversation and task submission while model/tool orchestration, state evolution, and artifact provenance remain server-side. This creates a controlled platform for studying capability transfer, multi-agent coordination, memory, and recovery, with reuse across different task workflows and future tasks.

Selected Projects

Nexgent: Scientific Agent Harness *Independent Project* **May 2026 – Present**

- **Problem:** Long-running scientific tasks need more than a chat interface: agents must connect to domain tools, coordinate specialized work, survive interruptions, expose risky actions, and leave a traceable experimental record. Nexgent was developed as the Agent Harness for prototyping these requirements in the AI + Fusion Energy project.
- **System:** Built a model-agnostic, frontend-neutral runtime shared by a PyQt6 desktop workspace, Textual TUI, and headless CLI; integrated 38 tools, automatic MCP tool bridging, multi-model routing, and sub-agent/workflow orchestration in pipeline, parallel, and phased modes.
- **Workflow functions:** Implemented progressive context compression, typed project/reference memory, session save–resume–fork, checkpoint rollback, permission gates, lifecycle hooks, goals, and background tasks for controlled and recoverable experiment execution.

oh-my-opencode (now oh-my-openagent) *Open-Source Contributor* **May 2026 – Present**

- **Research question:** Investigated which agent behaviors should remain invariant across OpenCode, Codex, Pi, and Claude Code rather than being entangled with one Harness. Proposed a downward-only Core–Adapter DAG separating reusable policy from Hook, provider, and UI effects.
- **Semantic abstractions:** Defined agent configuration as a zero-dependency core and model fallback as a session-scoped `idle/armed/exhausted` protocol with injected reachability and logging, turning implicit error handling into explicit, portable, and testable state transitions.
- **System validation:** Preserved existing call sites through re-export shims and tested cleanup, re-arming, chain exhaustion, and session isolation; continue to help lead the monorepo refactor, putting reusable Harness-independent policies into practice through plugins.

Where Is My Codex (Flow Rescue) *Course Project* **Spring 2026**

- **Physics:** Co-developed a 2D fluid-puzzle game and implemented Position-Based Fluids in Python/Taichi with uniform-grid neighbor search, density constraints, terrain collisions, and GPU/CPU backends.
- **Engineering:** Built five playable levels and a Tiled import pipeline; added headless tests and profiling for digging, collision, channel stability, goal collection, grid construction, and constraint solving.

Tiny PyTorch *Course Project* **Fall 2024**

- **Framework:** Built a PyTorch-style tensor and reverse-topological autodiff engine with core operators, fully connected layers, convolution, pooling, softmax loss, and end-to-end MLP/CNN training.
- **Acceleration:** Implemented C++/CUDA storage, transfers, custom kernels, and cuBLAS-backed computation with Python bindings; trained on MNIST and compared single-GPU and DataParallel workflows.

Honors & Awards

- National Scholarship and Merit Student, Peking University (2025)
- Second Prize, Beijing Division, National College Student Mathematical Modeling Competition (2025)
- Outstanding Learning Award and Third Scholarship, Peking University (2024)
- First Prize, Beijing College Student Mathematics Competition (2024)

Skills

Programming: Python, C/C++, CUDA, MATLAB, LaTeX, CMake, Taichi
Tools & Frameworks: PyTorch, PyQt6, Git, Linux, Scikit-learn, Jupyter, OpenFOAM
Languages: Chinese (native); English (CET-6: 613)